ELSEVIER

Precambrian Research 69 (1994) 157–167

Precambrian Research

Paleomagnetism of Precambrian dike swarms in the Harohalli area, south of Bangalore, India

E.M. Dawson*, R.B. Hargraves

Department of Geological and Geophysical Sciences, Princeton University, Princeton, NJ 08544, USA

Received June 15, 1992; revised version accepted January 27, 1994

Abstract

Seventy-two oriented core samples from twelve sites in three dike swarms in the Harohalli area, two of which have been dated radiometrically (Ikramuddin and Stueber, 1976) have been studied. Two old metadolerite dikes were completely unstable and gave no consistent results. After detailed AF and thermal demagnetization, and principal component analysis, three sites in E–W quartz dolerite dikes in the group dated at 2370 ± 230 Ma gave consistent vectors: $N=3$, $D=79.6$, $I=-82.3$, $k=191$, $\alpha_{95}=9.0$; pole 9.5°S, 242.4°C. One N.S. dike from this group is roughly reversed to the above. The fifth dike, an olivine dolerite, gave a completely distinct vector. Two of five alkali syenite dikes, dated at 814 ± 34 Ma gave a consistent vector: $N=2$, $D=26.9$, $I=83.0$, $k=1910$, $\alpha_{95}=5.7$, pole 24.75°N, 264.1°E.

The three E–W dolerite dike vectors correlate well with results from three other published dike studies, the mean of thirteen site vectors (presumed to date from ~2.4 Ga), yielding a pole at lat. 15.2°N, long. 55.6°E, $\alpha_{95}=10.2$.

1. Introduction

The Indian Peninsular shield is trellised by several generations of dolerite dikes, particularly in the granite–greenstone terrains (see Halls, 1982). Because they are so amenable to paleomagnetic study, they are the source of the vast majority of Precambrian paleomagnetic data from India, especially from the Dharwar craton (Naqvi et al., 1974). Unfortunately, dolerite is difficult to date radiometrically, leaving the age of most Precambrian paleo-poles from India essentially unknown (Hargraves and Bhalla, 1983).

Three distinct generations of dikes of different petrographic types occur in the granite–greenstone terrain in the Bidadi–Harohalli area of southeast Karnataka State (Fig. 1 from Ikramuddin, 1974). The Peninsular Gneiss and Closepet Granite here are cut by: (1) metamorphosed, but undeformed metadolerite and metanorite dikes; (2) a swarm of unmetamorphosed dolerite dikes; (3) a series of alkaline dikes. The petrology, geochemistry, and Rb–Sr geochronology of the unmetamorphosed dolerite and alkaline dikes have been studied by Ikramuddin (1974) and Ikramuddin and Stueber (1976), from whom the following descriptions are taken.

The unmetamorphosed dike suite includes olivine dolerite, normal dolerite, and quartz dolerite. On the basis of major and trace element abundances, Ikramuddin (1974) concluded that

* Present address: 900 W. 25th St. 3, Minneapolis, MN 55405, USA.

0301-9268/94/$07.00 © 1994 Elsevier Science B.V. All rights reserved
SSDI 0301-9268(94)00019-N

these represent various stages in the fractional crystallization of a single olivine tholeite parent magma. The dikes usually have chilled margins which are generally unaltered, and vary in width from 1 to 15 m, and in length from 45 m to 16 km.

The alkaline dikes, which include solvsbergites, tinguaites, and bostonites (Ikramuddin, 1974) are composed almost entirely of alkali feldspars and alkalic pyroxenes. Chemically, mineralogically, and texturally they resemble trachytes and phonolites. Chilled borders are rare, but the dikes are very fine grained across their entire widths, which range from 1 to 15 m; their lengths range from 90 m to more than 1 km.

As reported by Ikramuddin and Stueber (1976) seven whole-rock samples of the dolerite dikes yielded a Rb–Sr isochron age of 2370 ± 230 Ma, and eight whole-rock samples of the alkaline dike suite gave a Rb–Sr isochron age of 814 ± 34 Ma. The age, as determined by K–Ar method, of one sample from an alkaline dike (810 ± 25 Ma) agrees closely with this Rb–Sr age (Rb–Sr ages have been recalculated with $\lambda = 1.42 \times 10^{-11}$ yr^{-1}). The following paper is an abbreviated version of a thesis by Dawson (1984) which should be consulted for further details.

2. Sampling and laboratory procedure

Five sites in each of the alkaline and dolerite swarms plus two sites in metadolerites were sampled for paleomagnetic study by Hargraves in 1981. Six samples (blocks or cores drilled in the field) were gathered at each of the twelve sites, and oriented with sun and magnetic compasses. NRM measurements were made with a Schonstedt DSM-1 spinner magnetometer interfaced to an Apple IIe computer. Specimens were thermally demagnetized with a Schonstedt TSD-1 apparatus. The AF demagnetization equipment (maximum field 120 mT) rotates specimens around two perpendicular axes in an ambient DC field of less than 20 nT. All this equipment is housed in a magnetically shielded room.

Demagnetization data were analyzed using Zijderveld vector diagrams (Dunlop, 1979) with linear regression (Kirschvink, 1980) and using the converging remagnetization circles technique of Halls (1978). Vector means were calculated with both Fisher statistics and Bingham–Onstott statistics (Onstott, 1980).

The remanent magnetism in the two metadolerite dikes (sites 22, 23) was found to be extremely weak and unstable. No consistent results were obtained, and these sites will not be discussed further.

3. Dolerite dikes

The dolerite dike sample collection included representatives of the three petrographic types of Ikramuddin (1974): quartz dolerite, olivine dolerite and normal dolerite. As these give different paleomagnetic results (Table 1), they will be described separately.

Table 1

Site	N/n	Decl.	Inc.	K	α_{95}	Pole lat.	Pole long.
12	6/6	135.1	−84.7	110	6.4	19.6°S	250.7°E
14	6/6	246.9	−35.9	13	19.3	24.9°S	330.2°E
15	6/6	72.6	−78.1	84	7.4	5.2°S	235.6°E
16	5/6	60.3	−81.2	109	7.4	4.4°S	243.0°E
17	5/6	71.8	69.4	–	3.4,4.7	20.9°S	295.1°E
Harohalli quartz dolerite dike mean (sites 12, 15 and 16):							
	N=3	79.6	−82.3	191	9.0	9.5°S	242.4°E
Mean of sites 12, 15 and 16, with site 17 (dolerite) reversed:							
	N=4	112.1	−88.7	30	17.1	13.4°S	254.4°E
Alkaline dikes							
18	5/6	18.4	83.9	65	9.5	23.8°W	261.6°E
21	6/6	32.0	82.0	28	12.8	25.6°S	266.8°E
Harohalli alkaline dike mean (2 sites):							
	N=2	26.9	83.0	1910	5.7	24.7°S	264.1°E

Note: N/n signifies the number of samples used in calculation/number of samples measured. The converging remagnetization circle technique of Halls (1978) was used for site 17. The intersection of the great circles was calculated with Bingham–Onstott statistics (Onstott, 1980) which provides two perpendicular α_{95}-values.

3.1. Quartz dolerite

Sites 12, 15 and 16 (see Fig. 1) were collected from three E–W-striking quartz dolerite dikes. Petrographically these are exactly as described by Ikramuddin (1974): the mineral assemblage is oligoclase–andesine, augite (ferroaugite), with subsidiary quartz, alkali feldspar, amphibole, and micropegmatite. The rocks are relatively coarse grained and have a poikilitic texture. As seen in polished thin section, the opaque minerals are titano-magnetite and disseminated sulfides. Magnetite grains are subhedral to skeletal and display fine trellis "oxy-exsolution" lamellae of ilmenite as well as blocky sandwich lamellae. Pyrite, and what appears to be chalcopyrite, are both present. In site 12 samples, some primary hemo-ilmenite grains contain basal plane exsolution lamellae of titano-hematite. No hemo-ilmenite was seen in samples from site 15 or 16.

Considering their great age, these rocks are remarkably fresh, and alteration, where present, is judged most likely to be deuteric. Plagioclase, and more frequently the subsidiary potassium feldspar, are partly sericitized. Clinopyroxene crystals are often altered around their edges to uralite, a green amphibole. Some titano-magnetite grains are surrounded by, and in a few cases intergrown with, dark, reddish-brown biotite.

NRM directions of the eighteen samples from sites 12, 15, and 16 are scattered, with a slight grouping to the northwest with shallow negative inclinations. Two NRMs are antipolar to this direction. AF demagnetization (see Figs. 2, 3) removes a scattered, large, soft component by 20 mT. From 20 to 80 mT a steep negative component, which we believe to be the primary remanence, is progressively removed in all but one of the samples. In sites 12 and 15, the linear magnetization trajectory through the data is slightly oblique to the origin, but upon further demagnetization up to the limit of our apparatus (120

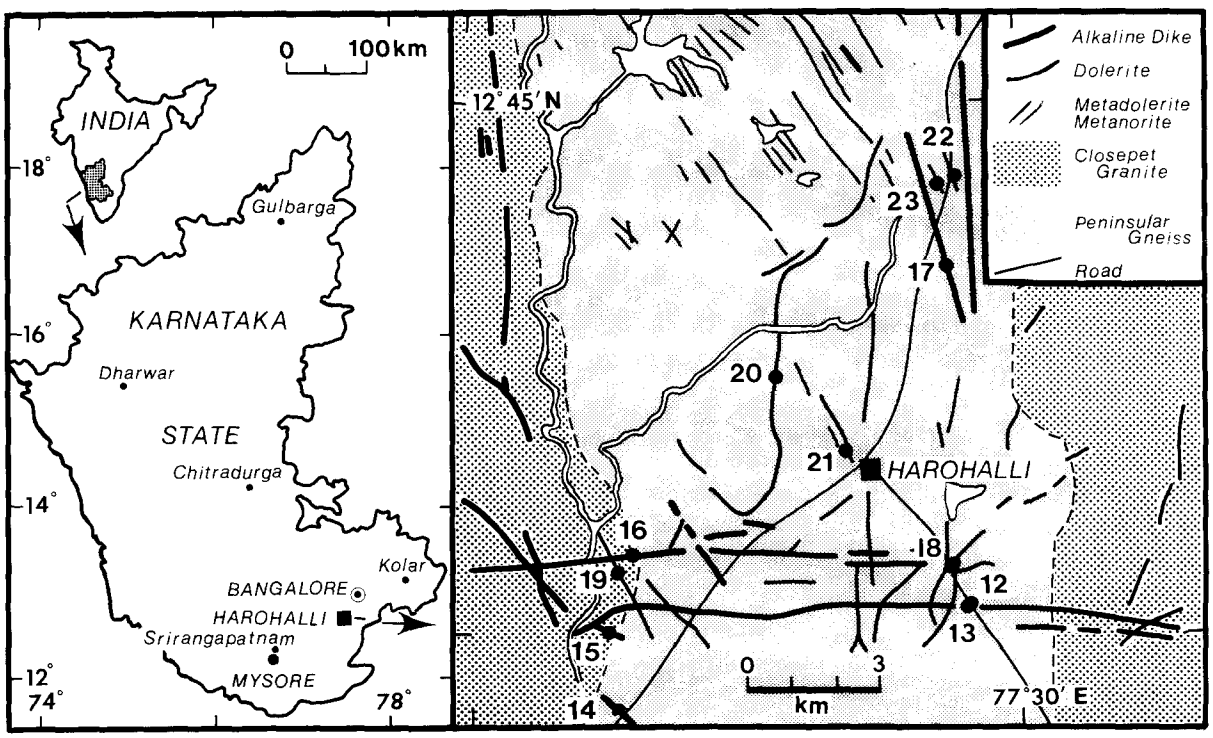


Fig. 1. Geologic map showing location of Harohalli, and metadolerite, dolerite, and alkalic dikes. Sampling sites are numbered. After Ikramuddin (1974).

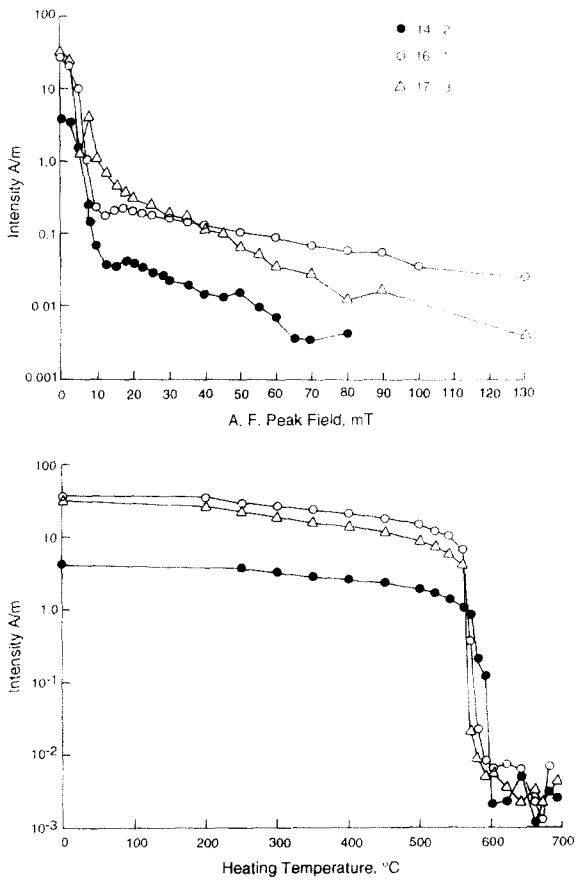


Fig. 2. AF and thermal demagnetization curves for representative samples from site 16 (quartz dolerite), site 14 (olivine dolerite), site (dolerite dikes).

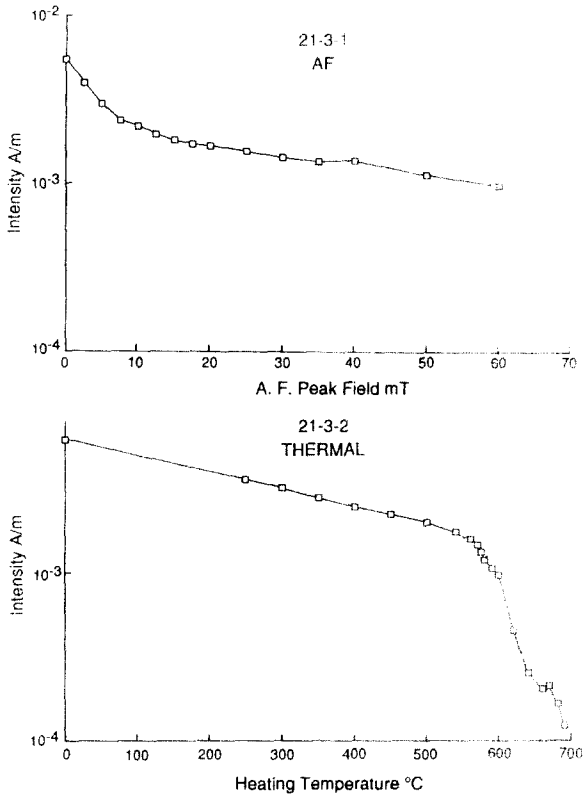


Fig. 3. Representative Zijderveld plots showing migration of vectors during AF and thermal demagnetization of quartz dolerite dike samples from sites 12 and 16.

mT) it begins to bend toward the origin. The significance of this small, high-coercivity component is not clear.

Thermal demagnetization (Figs. 2, 3) of one pilot specimen from each site isolated the steep negative component only in site 12. Here, a southeast shallow positive component was removed up to 540°C, with the steep negative component surviving between 540°C and 620°C. While the steep negative vector persists up to 620°C at site 12, the remanence was largely destroyed by 580°C at sites 15 and 16. This difference in blocking temperatures is entirely consistent with the magnetic mineralogy observed in polished thin section in that only samples from site 12 contain hemo-ilmenite (with titano-hematite lamellae) in addition to the magnetite.

3.2. Olivine dolerite

Site 14 was collected from an olivine dolerite dike 10 to 13 m wide striking NW–SE (Fig. 1). The primary mineral assemblage includes plagioclase, orthopyroxene, clinopyroxene, olivine, with minor opaques; the rock has an ophitic texture. In thin section, it does not have the fresh appearance of samples from the quartz dolerites. While the feldspars show little alteration, both pyroxenes and olivines are extensively serpentinized. In addition, clinopyroxenes are rimmed with uralite, while olivine grains have thin reaction rims rich in magnetite. Primary titano-magnetite grains are subhedral with broad ilmenite lamellae, and contain inclusions of sulfide min-

erals. In many grains the magnetite has been extensively altered, presumably to an intergrowth of hematite and rutile, leaving only the ilmenite trellis lamellae unaltered.

AF treatment up to 15 mT removed a soft, viscous component of inconsistent orientation, reducing the intensity of magnetization by two orders of magnitude. From 15 to 80 mT a southwest, negative component was removed from all six samples, the mean orientation of which is $D=246.9$, $I=-35.9$, $\alpha_{95}=19.3$. Thermal demagnetization of a pilot specimen was dominated by the large viscous component which persisted up to 590°C. Remanence was not completely destroyed until between 590 and 600°C, but considering uncertainties in the furnace temperature and the exact Curie point of magnetite, we believe that this does not indicate the presence of a magnetic carrier other than pure magnetite. While the unstable remanence is most probably carried by the large multidomain magnetite grains, the stable component could well be associated with fine single-domain (or PSD) magnetite particles produced by the deuteric alteration.

3.3. Dolerite

Site 17 (Fig. 1) was collected from a dolerite dike 3 m wide with a strike slightly west of due north. The rock is composed almost entirely of plagioclase and clinopyroxene with four to five modal percent magnetite, abundant sericite and minor uralite, and lacking quartz or olivine. The texture is ophitic to intergranular. Titano-magnetite grains are skeletal, together with fine blotchy ilmenite exsolution, together with large magnetite patches. Long fresh sulfide veins are present along silicate cleavages.

NRM directions were scattered and fairly resistant to both AF and thermal demagnetization. This site, located along a ridge crest, could well have been struck by lightning repeatedly. During AF demagnetization the scattered NRM directions initially begin to converge on a steep positive vector, but by 20 mT the specimens' declination and inclination has started to move back toward their NRM directions. Zijderveld plots displayed a distinct S-shape. The low field and high field portions of the plot are parallel but are offset by an intermediate coercivity component. The random NRM component apparently has a very broad range of coercivity and is removed throughout the AF demagnetization process, while another component has a more restricted coercivity range concentrated between 20 and 50 mT. As the demagnetization data showed streaking, the converging remagnetization circles analysis technique of Halls (1978) was employed.

4. Alkaline dikes

Oriented samples were collected from each of five alkaline dikes (sites 13, 18, 19, 20, 21). Petrographically, these are as described by Ikramuddin (1974), except that we note a pervasive alteration of the rock at three of the sites (18, 19 and 21). In hand samples the altered dike rocks have a reddishbrown color in contrast to the grey/green color of the unaltered material. Polished thin sections of the former revealed a diffuse hematite staining and alteration of feldspars and the cores of pyroxene grains. The alteration is most likely deuteric, produced when these shallow, subvolcanic intrusions encountered meteoric water during emplacement.

Rocks from the two unaltered sites (13 and 20) contain almost no opaque minerals. Remanence in these was weak and scattered and dropped below the sensitivity of the Schonstedt spinner after only a few demagnetization steps. No further investigation was possible. Fortunately, remanence intensities of samples from the altered sites (18, 19 and 21) were one or two orders of magnitude greater, well within the sensitivity range of the magnetometer. Response to AF and thermal demagnetization of cores from site 21 are illustrated in Figs. 4 and 5.

The site 21 dike is essentially a trachyte, consisting almost entirely of alkali feldspar which has been pervasively altered, and permeated with small specks of hematite. A fibrous yellow chlorite replaces pre-existing pyroxenes. Hematite is also present as feathery cryptocrystalline aggregates often pseudomorphic after a cubic mineral,

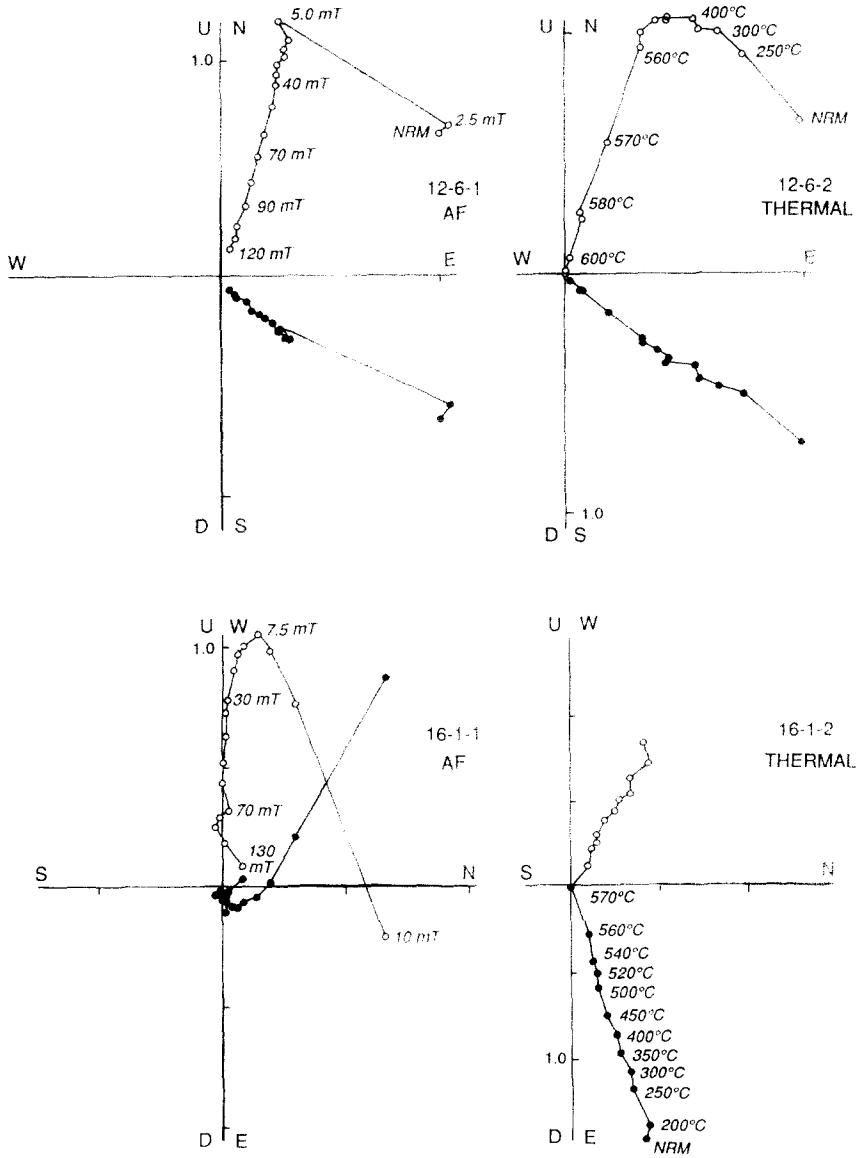


Fig. 4. AF and thermal demagnetization curves for alkaline dike samples from site 21.

possibly pyrite. Thus, hematite is present in a spectrum of grain sizes. Thermal demagnetization, however, revealed the presence of a distinct component with a 590°C blocking temperature, suggesting the presence,in addition,of magnetite. NRM intensity averaged about 5×10^{-3} A/m, with due north declinations and shallow negative inclinations. At low fields, AF demagnetization removed a soft component parallel to the present-day Earth's field, as well as an easterly component in some specimens. Above 30 mT a spurious rotational remanent magnetization (RRM) became increasingly troublesome, causing directional instability in these samples. Thermal demagnetization, on the other hand, erased the due north negative component by 250 to 300°C. Further heating eliminated other random secondary components, but in the range of

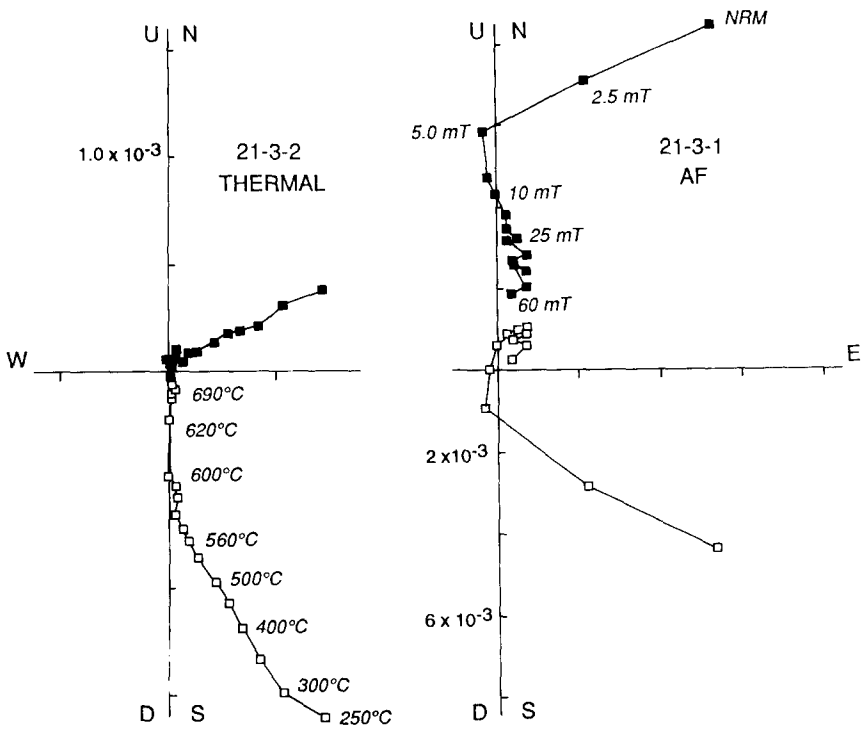


Fig. 5. Zijderveld plots showing migration of vectors during AF and thermal demagnetization of core from site 21.

590–670°C a consistent, vertical, positive component was removed from all six specimens ($D=32.9, I=82.0, \alpha_{95}=12.8$).

In site 18 samples the feldspars are relatively fresh, while the abundant pyroxenes are intensively altered to a fine-grained cryptocrystalline hematite-rich aggregate. Thermal demagnetization again indicates that magnetite is present and this phase evidently carries a greater portion of the total remanence than in site 21. NRM intensities average around 5×10^{-2} A/m, an order of magnitude greater than those of site 21. NRM directions were toward the east with shallow inclinations, and due south with positive inclinations (antipolar to site 21 NRMs). AF demagnetization erased the NRM directions by 10 to 15 mT. Thereafter, in half the samples, a steep positive component similar to the stable vector in site 21 was removed, while the other three samples were dominated by a hard northerly negative component, similar to the NRM found in site 21. RRM, again, became a problem above 30 mT. During thermal demagnetization, however, NRM directions persisted up to 680°C.

We observed that for site 21 the northerly negative component was resistant to AF demagnetization, but was removed at low temperatures during thermal demagnetization. The various NRM components of site 18, on the other hand, were soft to AF demagnetization but resistant to heating. In an attempt to isolate the steep positive component from all of the site 18 samples, we used a combination of thermal and AF demagnetization, whereby specimens were heated stepwise to 400°C to remove the northerly negative component and were then AF-demagnetized to remove the southerly and easterly overprints. For two "anomalous" samples, this technique was fairly successful. The steep component was isolated between 20 and 45 mT, although it did not decay precisely toward the origin. Combining the three sample vectors isolated by AF demagnetization with the two provided by thermal–AF demagnetization, we obtain a mean site vector close to that of site 21 ($D=18.8$, $I=84.0$, $\alpha_{95}=9.5$).

Site 19, the third 'altered' dike, is coarser grained than the others, and contains a greater modal abundance of pyroxenes. NRM intensities were comparable to those of site 18, but directions were more scattered. Thermal, AF, and thermal–AF demagnetization were of no use in isolating any consistent vectors. Thus, only two of the five alkaline dikes sampled provided useful paleomagnetic information.

The Precambrian age of the alkaline dikes and the presence within them of magnetite, together with hematite in a wide range of particle grain-size all probably contribute to the complex, multi-component magnetizations observed. The most consistent secondary component is northerly negative. With a few notable exceptions, this is removed at low temperatures but is resistant to AF demagnetization. Our best estimate of its orientation ($D=8.3$, $I=-37.3$, $\alpha_{95}=5.5$) is provided by site 21 (diverging) remagnetization circles in the range of 0–400°C. The origin of this component is not clear; it could be a PTRM associated with the Early Tertiary Deccan Traps which give similar remanence vectors, but no such overpoint has been seen in other studies in this area (T. Radhakrishna, pers. commun, 1992). The easterly component present in many samples was more difficult to isolate and its origin is less clear. While it is removed by low AF fields, it persists to high temperatures.

We believe that the steep component most likely represents a primary remanence carried by hematite produced during deuteric alteration. It is resistant to both thermal and AF demagnetization and is present in all but one sample from sites 18 and 21.

5. Discussion

Dolerite dikes

The site mean directions obtained from the five dolerite dikes are shown in Fig. 6 and listed in Table 1.

Whereas the mean vectors from sites 12, 16 and 17, all east–west striking quartz dolerite dikes, give very similar steeply positive inclination vectors, those from site 14 (olivine dolerite, striking NW) and site 17, (normal dolerite, striking N–S) are significantly displaced.

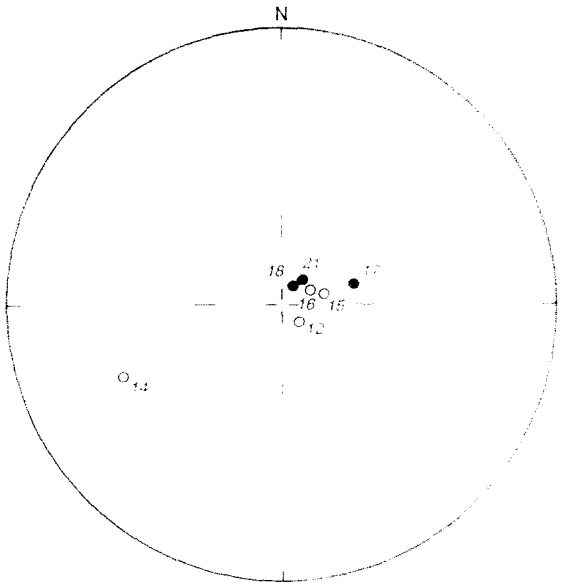


Fig. 6. Harohalli site vectors.

The consistency in the AF (and thermal) demagnetization response and orientation of the resulting vectors for 17 (out of 18) samples from sites 12, 15 and 16 encourages the conclusion that this is a characteristic remanence. The rocks are extremely fresh and their Fe–Ti oxides are pristine. Even without a contact test,there is no reason to doubt that this stable remanent magnetization dates from the time of intrusion and cooling of the dikes, and the mean yields a useful paleomagnetic pole.

The olivine dolerite of site 14 is distinctly more altered, and the history of its remanent magnetization is much less clear. Furthermore its RM's are more scattered ($k=13$), and the mean direction is unique (see Table 1). The dependability and significance of this site vector is marginal.

Petrographically, the dolerite dike samples from site 17 (strike northerly), are also fresh with little sign of alteration of silicates or Fe–Ti oxides. Lightning-induced secondary components are the likely cause of the overprinted magnetizations. After demagnetization the derived site vector is within 30° of being reversed with re-

spect to the mean of sites 12, 13 and 16, but whether it should be included within the latter group is not clear. Further field and paleomagnetic studies of more dikes in the Harohalli swarm are required to clarify the relationships between them. The mean of the three Harohalli quartz dolerite dike sites, however, is considered to yield a good paleomagnetic pole.

All of the dikes sampled in this study were judged to be consanguineous by Ikramuddin and Stueber (1976), who plotted Rb–Sr data from representatives of all petrological types in their isochron plot, yielding an errorchron age of 2420 ± 246 m.y. The large age uncertainty permits speculation that the distinct magnetizations of the different petrologic types might be due to significant age differences. An isochron plot of the Rb/Sr data from east–west dikes alone yields an age of 2.21 Ga, but because there are only three data points, the errors cannot be constrained. In any case, this revised age is within the large error limits of the original report.

The age of the Harohalli quartz dolerite pole is thus not well established, despite the radiometric dating which has been performed. Our best estimate at present is that it is "around" 2.2 Ga.

5.2. Alkaline dikes

Only two of the five dikes sampled yielded consistent data. These were intensely altered hydrothermally, and their remance was clearly acquired as a result of this alteration. Assuming the Rb–Sr whole-rock age reflects the time of intrusion, the concordant K–Ar ages rule out any major post-emplacement thermal event capable of completely remagnetizing them. The alteration is therefore inferred to be deuteric, and the mean vector from two sites relates to the 800 m.y. age.

5.3. Similarity between dolerite and alkaline dike RM orientations

As evident from Table 1, the mean remanent vectors from these two dike swarms are both very steep, but whereas the quartz dolerites (sites 12, 15 and 16) have positive inclinations, the alkaline dikes are negative. Dolerite dike site 17 also has positive inclination, and this similarity to the alkaline dikes further detracts from its reliability. We believe, however, that notwithstanding the similar steep inclinations of the quartz dolerites and alkaline dikes (and the resulting poles), the contrast in polarity precludes remagnetization of the older dolerites (with possible exception of site 17) at the time of intrusion of the younger alkaline dikes.

6. Comparison with other Precambrian poles from the Dharwar craton

Hargraves and Bhalla (1983) reviewed all Precambrian paleomagnetic data from India published prior to that year, the bulk of which from the Dharwar craton was obtained from the numerous dike swarms which cut the subcontinent (Halls, 1982). At that time (1983), none of the dikes for which paleomagnetic data were available had been studied radiometrically. Conversely, the Harohalli dikes which had been dated (Ikramuddin and Stueber, 1976), had not been studied paleomagnetically. This study (Dawson, 1984) undertook to rectify that situation.

Since 1983, numerous K–Ar dates for dikes have been reported by Bhattacharji (1987) and Murthy et al. (1987). The results (14 dates) show a considerable spread between ~2.2 Ga and 0.8 Ga. No clear definition of swarms is evident. Thus the ages of the five paleomagnetic vector groups discerned by Hargraves and Bhalla (1983) remain poorly constrained. In addition, paleomagnetic studies of several other dikes in the Dharwar craton have been published (Kumar and Bhalla, 1983; Subba Rao and Radhakrishna Murthy, 1985; Radhakrishna et al., 1986; Venkatesh et al., 1987).

The results obtained in this study from the dolerite dikes are summarized in Fig. 5. The vector for site 14 is unlike any of the previously published dike vectors from the Dharwar craton from metamorphosed banded quartz magnetite. Crawford (1969) reported a whole-rock Rb–Sr

age of 2590 ± 40 Ma for these gneisses. With only one site (14) represented, however, the significance of this distinct vector is uncertain.

Sites 12, 15 and 16, all from E–W dikes form an excellent group, with site 17 (striking. NNW) possibly being reversed. These three (12, 15 and 16) correlate closely with Group V of Hargraves and Bhalla (1983), which was based on studies by Hasnain and Qureshy (1971) and Bhalla and Rao (1980). In 1983, Kumar and Bhalla published results from a series of dikes 150 km north of Harohalli. Of the five dolerite dikes (thirteen sites) they studied, they report five site vectors from two separate dikes (striking ENE) which correspond well with the present results ($D=57°$, $I=-69°$, $\alpha_{95}=7°$). None of the other newer studies report vectors which could be correlated with this high inclination group.

The mean of the thirteen site poles resulting from these four studies yields a northern hemisphere pole at lat. 15.3°N, long. 54.3°E, $\alpha_{95}=10.7$. Assuming dikes in this expanded Group V (Hargraves and Bhalla, 1983) to be penecontemporaneous, then this pole is approximately dated by the results of Ikramuddin and Stueber (1976) for the Harohalli swarm (2370 ± 230 Ma).

Only two alkaline dikes sites gave consistent results (Table 1, pole lat. 15.2°N, long. 55.6°E, $\alpha_{95}=10.2$), but the swarm is well dated by Ikramuddin and Stueber at 814 ± 34 ma. The Malani rhyolites (740 Ma) are closest in age, yet the well determined Malani rhyolite pole (Klootwijk, 1975, 80.5°N, 43.5°E, dp=8, dm=11.5) is, as might be expected, significantly removed.

7. Conclusions

While the "Harohalli" quartz dolerite pole is relatively well determined paleomagnetically ($N=3$ sites), its 2.2 Ga estimated age has large uncertainty. This age uncertainty applies at least as much to the calculated Group V pole in which the Harohalli data are included. For this reason alone, comparisons with available results from other Indian cratons (let alone comparisons with other continents) are considered premature and will not be attempted.

Acknowledgments

The samples used in the study were collected by Hargraves during tenure of an Indo–US Subcommission Fellowship (1980–1981) at the Nadonal Geophysical Research Institute, Hyderabad. The support of Dr. Balakrishna (then Director) and Dr. M.S. Bhalla is gratefully acknowledged. The laboratory at Princeton where the measurements were made, was supported by the N.S.F.

References

Bhalla, M.S. and Hansraj Rao, N.J.V.P., 1980. Paleomagnetic studies of Precambrian dikes from Hunsar, Holenarsipur and Tiptur in Karnataka, India. Geoview 5: 157–172.

Bhattacharji, S., 1987. Lineaments and igneous episodes in the evolution of the Intracratonic Proterozoic basins on the Indian shield: Recent Res. Geol., 13: 1–15.

Crawford, A.R., 1969. Reconnaissance Rb–Sr dating of the Precambrian rocks of Southern Peninsular India. J. Geol. Soc. India, 7: 91–110.

Dawson, E.M., 1984. Paleomagnetism of the Bidadi–Harohalli Dike swarm. Senior Thesis, Dept. of Geol. and Geophys. Sciences, Princeton Univ., Princeton, NJ.

Dunlop, D.J., 1979. On the use of Zijderveld vector diagrams in multicomponent paleomagnetic studies. Phys. Earth Planet. Int., 20: 12–24.

Halls, H.C., 1978. The use of converging remagnetization circles in paleomagnetism. Phys. Earth Planet. Int., 16: 1–11.

Halls, H.C., 1982. The importance and potential of mafic dike swarms in studies of geodynamic processes. Geosci. Can., 9: 145–154.

Hargraves, R.B. and Bhalla, M.S., 1983. Precambrian paleomagnetism in India through 1982: a review. In: S.M. Naqvi and J.J.W. Rogers (Editors), Precambrian of South India. Mem., Geol. Soc. India, 4: 491–524.

Hasnain, I. and Qureshy, M.N., 1971. Paleomagnetism and geochemistry of some dikes in Mysore State, India. J. Geophys. Res., 76: 4786–4795.

Ikramuddin, M., 1974. Geochemistry and Geochronology of Precambrian Dikes from Mysore State, India. Unpubl. Ph.D. thesis, Karnatak Univ.

Ikramuddin, M. and Stueber, A.M., 1976. Rb–Sr ages of Precambrian dolerite and alkaline dikes, southeast Mysore State, India. Lithos, 9: 235–241.

Kirschvink, J., 1980. The least-squares line and plane and the analysis of paleomagnetic data, Geophys. J. R. Astron. Soc., 62; 699–718.

Klootwijk, C.T., 1975. A note on the paleomagnetism of the late Precambrian Malani Rhyolites near Jodhpur, India. J. Geophys., 41; 189–200.

Kumar, A. and Bhalla, M.S., 1983. Paleomagnetics and igneous activity of the area adjoining the south-western margin of the Cuddapah Basin, India. Geophys. J. R. Astron. Soc., 373: 27–37.

Murthy, Y.G.K., Babu Rao, V., Guptasarma, D., Rao, J.M., Rao, M.N. and Bhattacharji, S., 1987. Tectonic, petrochemical and geophysical studies of mafic dike swarms around the Proterozoic Cuddapah Basin, South India, In: H.C. Halls and W.F. Fahrig (Editors), Mafic Dike Swarms, Geol. Assoc. Can. Spec. Pap., 34: 303–316.

Naqvi, S.M., Divakara Rao, V. and Narain, H., 1974. The protocontinental growth of the Indian shield and the antiquity of its rift valleys. Precambrian Res., 1: 345–348.

Onstott, T.C., 1980. Application of the Bingham distribution function in paleomagnetic studies. J. Geophys. Res., 85: 1500–1510.

Radhakrishna, T., Poornachandra Rao, G.V.S., Mitchell, J.G. and Venkatesh, A.S., 1986. Proterozoic basic dike activity in Kerala along the western continental margin of India. J. Geol. Soc. India, 27: 244–253.

Subba Rao, Y.V. and Radhakrishna Murthy, 1985. Paleomagnetism and age of dolerite dikes in Karimnagar district, Andhra Pradesh, India. Geophys. J. R. Astron. Soc., 82: 331–337.

Venkatesh, A.S., Poornachandra Rao, G.V.S., Prasada Rao, N.J.V. and Bhalla, M.S., 1987. Paleomagnetic and geochemical studies on dolerite dikes from Tamil Nadu, India. Precambrian Res., 34: 291–310.